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National Electric Power Regulatory Authority (NEPRA)

NOTIFICATION

Islamabad, the 1st day of September, 2015

S.R.O 892 (1)/2015.— In exercise of the powers conferred by section 47 read with section 7 (1) of the Regulation of Generation, Transmission and Distribution of Electric Power Act, 1997 (XL of 1997), the National Electric Power Regulatory Authority, is pleased to make the following Regulations to establish a framework for the regulation of Distributed Generation by using alternative and renewable energy and net metering, namely: -

1. Short title, commencement and application. — (1)These Regulations may be called the National Electric Power Regulatory Authority (Alternative & Renewable Energy) Distributed Generation and Net Metering Regulations, 2015.

(2) They shall come into force at once.

2. Definitions. - (1) In these Regulations unless there is anything repugnant in the subject or context,—

(a)"Act" means the Regulation of Generation, Transmission and Distribution of Electric Power Act, 1997 (XL of 1997)

(b) "Agreement" means the agreement between the Distribution Company and the Distributed Generator on the Authority's approved format as per Schedule I;

(c)"Applicable Documents" means the rules and regulation issued in pursuance of the Act by the Authority, from time to time, the generation, distribution and transmission licences, the Grid and Distribution Codes and any documents, instruments, approvals, directions or authorizations issued or granted by the Authority in exercise of its powers under the Act and any document in each case of a binding nature applicable to the licensee;

(d) "Applicable Tariff" means tariff approved by the Authority for the relevant period and category of consumers of the Distribution Company;

(e)"Applicant" means a consumer of a Distribution Company which submits an Application to interconnect its Distributed Generation Facility to the Distribution System of the Distribution Company and who applies for grant of the license to operate a Distributed Generation Facility as a Distributed Generator;

(f) "Application" means the application submitted by an Applicant to Distribution Company, for interconnection of a Distributed Generation Facility to the Distribution System of a Distribution Company and includes application for grant of license to the Authority;

(g) "Authority" means the National Electric Power Regulatory Authority established under the Act;

(h) "Billing Cycle" means energy recorded by the meters in a period of thirty days;

(i) "Distributed Generation Facility" means a facility set up by a Distributed Generator using Solar or Wind energy resource for generation of electricity up to 1 MW;

(j) "Distributed Generation" means electrical power generation by solar or wind that is interconnected to the Distribution System of the Distribution Company at Interconnection Point;

(k) "Distributed Generator" means a Distribution Company's 3 phase 400V or 11kV consumer i.e. domestic, commercial or industrial and who owns and/or operates the Distribution Generation Facility, and is responsible for the rights and obligations related to the Agreement and licensed by the Authority under these Regulations;

(l) "Distribution System" means the distribution facilities situated within the Service Territory owned or operated by the licensee for distribution of electric power including, without limitation, electric lines or circuits, electric plant, meters, interconnection facilities or other facilities operating at the distribution voltage, and shall also include any other electric lines, circuits, transformers, sub-stations, electric plant, interconnection facilities or other facilities determined by the Authority as forming part of the distribution system, whether or not operating at the distribution voltage;

(m) "Fault" means an equipment failure, conductor failure, short circuit, or other condition resulting from abnormally high or low amounts of current from the power system;

(n) "Grid Code" means the guidelines, standards and procedures of technical and commercial aspects for the access, use and operation of transmission system and transmission facilities of NTDC as modified and approved by Authority from time to time;

(o) "Interconnection Facilities" means the equipment, including, without limitation, electrical lines or circuits, transformers, switch gear, safety and protective devices, meters or electrical plant, used for interconnection services;

(p) "Interconnection Point" means the point where the metering, installation and protection apparatus of the Distributed Generator is connected to the Distribution System of the Distribution Company;

(q) "kWh" means kilowatt hour;

(r) "MW" means megawatt;

(s) "Net Energy Billing" means a billing and metering practice under which a Distributed Generator is billed on the basis of net energy over the billing cycle;

(t) "Net Energy" means a balance (positive or negative) of the kWh generated by Distributed Generator against the kWh supplied by Distribution Company at the end of Billing Cycle;

(u) "Net Metering Facility" means a facility comprising of one or two meters for measuring the kWh generated by Distributed Generator and supplied by Distribution Company for determining the net energy;

(v) "Registrar" means a person designated by the Authority to register and record the receipt of communications, applications and petitions filed with the Authority and to perform such other duties under these Regulations as may from time to time be assigned by the Authority; and

(w) "Tariff" means the rates, charges, terms and conditions for sale of electric power to consumers as approved by the Authority and duly notified by the Federal Government from time to time.

2) The words and expressions used but not defined in these Regulations shall have the same meaning as are assigned to them in the Act.

APPLICATION AND INTERCONNECTION PROCESS

3. Application Process for Interconnecting Distributed Generation Facility.—

(1) Any person who meets the requirements of a Distributed Generator as defined under the regulations 2(k) is eligible for submitting application as specified in Schedule-II to a Distribution Company.

Provided that the Distribution Company shall be bound to provide information and Authority's approved documents in response to the request from Applicant free of cost within two working days.

(2) Application to Distribution Company along with necessary documents shall be submitted by intending Distributed Generator to Distribution Company.

(3) Within five working days of receiving an Application, the Distribution Company shall acknowledge its receipt and inform the Applicant whether the Application is completed in all respect.

Provided that in case of any missing information or documents the Applicant shall provide the same to Distribution Company within seven working days of being informed by Distribution Company.

(4) Upon being satisfied that the Application is complete in all respect, the Distribution Company shall perform an initial review to determine whether the Applicant qualifies for Interconnection Facility, or may qualify subject to additional requirements.

Provided that the initial review shall be completed within twenty working days.

(5) In case the initial review reveals that the proposed facility is not technically feasible, the Distribution Company shall return the Application and communicate the reasons to the Applicant within three working days after the completion of initial review.

(6) If the Distribution Company is satisfied that the Applicant qualifies as Distributed Generator, then the Distribution Company and the Applicant shall enter into an Agreement within ten working days and Distribution Company shall send a copy of the Agreement to the Authority within seven working days of the signing of the Agreement.

(7) Within seven working days of execution of the Agreement, the Distribution Company shall issue the Connection Charge Estimate to the Applicant for the proposed interconnection facility up to the Interconnection Point including the metering installation.

(8) The Applicant shall make the payment of Connection Charge Estimate within twenty days of its issuance.

(9) The Distribution Company shall install and commission the proposed interconnection facility within thirty days of the payment of demand notice by the Applicant.

Provided that the net metering arrangement shall commence upon grant of license to the Distributed Generator in accordance with Regulation 4 of these Regulations.

4. Licensing.– (1) Notwithstanding anything contained in NEPRA Licensing (Application and Modification Procedure) Regulations, 1999 any consumer who enters into an Agreement with the Distribution Company under net metering arrangement qualifies for grant of a Distributed Generator License.

(2) Distribution Company shall forward the Application for grant of License as specified in Schedule -III to the Authority along with following;

(a) Agreement
(b) Application for exemption from the requirement of section 24 of the Act as specified in Schedule-IV,
(c) Evidence of deposit of fee as may be specified by the Authority as specified in Schedule-V
(d) Affidavit by Distributed Generator as specified in Schedule-VI

(3) The Authority may, on receipt of the Application and the documents specified in sub-regulation (2), grant a license as specified in Schedule VII to the Applicant.

5. General Powers, Rights and Obligations of the Distribution Company.— (1) A Distribution Company shall,

(a) allow any of its consumers to establish Distributed Generating facilities to be interconnected with its Distribution System using either (a) a standard meter capable of registering the flow of electricity in two directions, or (b) two separate meters one for selling electricity to the Distribution Company and other for purchasing electricity from the Distribution Company.

(b) enter into an Agreement with the Distributed Generator, and shall grant interconnection approval under Regulation 3 (6) after following due process

Provided that the approval of interconnection facility shall not be unreasonably withheld.

(c) have the right to review the design of a Distributed Generation Facility and Interconnection Facilities and to inspect the same prior to the commencement of parallel operation with its Distribution System and may require the Distributed Generator to make modifications as necessary to comply with the requirements of these Regulations.

(2) A Distribution Company may limit the operation and/or disconnect or require the disconnection of a Distributed Generation Facility from its Distribution System at any time, with or without notice, in the event of Fault.

(3) A Distribution Company may also limit the operation and/or disconnect or require the disconnection of Distributed Generation Facility from its Distribution System upon the provision of thirty days written notice for the conditions which include as follows:

a) To allow for routine maintenance, repairs or modifications to the Distribution System of the Distribution Company;
b) Upon Distribution Company's determination that Distributed Generation Facility is not in compliance with these Regulations;
c) Upon termination of the Agreement.

6. Rights and Obligations of the Distributed Generator.— (1) A Distributed Generator shall operate and maintain its Distributed Generation Facility and Interconnection Facilities in accordance with prudent electrical practices.

(2) The Distributed Generator shall not have any right to utilize Distribution Company's Interconnection Facilities for the sale of electricity to any other person.

Terms of Agreement, Termination of Agreement and Dispute Resolution

7. Term of Agreement.— (1) The term of the Agreement between Distributed Generator and Distribution Company shall be three years with effect from commissioning of Distributed Generator Facility.

(2) At the expiry of initial term, the Agreement may be automatically renewed by the mutual understanding between the Distributed Generator and the Distribution Company for another term of three years and so on.

8. Termination of Agreement. (1) The Distributed Generator may terminate the Agreement upon thirty days written notice if the Distributed Generator decides to discontinue the sale of electricity to the Distribution Company.

(2) The Distribution Company shall not terminate the Agreement in any event without prior approval of the Authority.

(3) All rights and obligations accrued up to termination shall continue in force upon termination.

DISTRIBUTED GENERATION FCAILITY DESIGN AND OPERATING REQUIREMENTS

9. Protection Requirements.— (1) The protection and control diagrams for the interconnection of the Distributed Generator shall be in accordance with the provisions of the Grid and Distribution Codes and approved by the Distribution Company prior to commissioning of the proposed Interconnection Facilities and a typical single line diagram as specified in Schedule-VIII.

(2) The Distributed Generator shall be responsible for the installation of equipment, including, without limitation, electrical lines or circuits, transformers, switch gear, safety and protective devices, meters or electrical plant, to be used for interconnection.

Provided that, if the Distributed Generator is unable to install equipment, including, without limitation, electrical lines or circuits, transformers, switch gear, safety and protective devices, meters or electrical plant, used for interconnection, the Distribution Company may execute the requisite work in case the Distributed Generator offers to deposit the cost to be incurred on the requisite work at mutually agreed terms.

(3) The protective functions shall be equipped with automatic means to prevent reconnection of the Distributed Generation Facility with the Distribution facilities of the Distribution Company;

Provided that the service voltage and frequency is of specified setting and is stable and mutually agreed between the Distribution Company and the Distributed Generator.

(4) The Distributed Generator will furnish and install a manual disconnect device that has a visual break to isolate the Distributed Generation Facility from the Distribution facilities.

(5) The grid connected inverters and generators shall comply with Underwriter Laboratories UL 1741 standard (Inverters, Converters, Controllers and Interconnection System Equipment for Use with Distributed Energy Resources) which addresses the electrical interconnection design of various forms of generating equipment, IEEE 1547 2003, IEC 61215, EN or other international standards.

10. Prevention of Interference.- (1) The Distributed Generator shall not operate such equipment that superimposes upon the Distribution System a voltage or current that interferes with Distribution Company's operations, service to its consumers, or communication facilities.

(2) In the event of such interference, the Distributed Generator must diligently pursue and take corrective action at its own expense after being given notice and reasonable time to do so by the Distribution Company.

(3) On account of any failure on part of the Distributed Generator to take timely corrective action, the Distribution Company may, without liability, disconnect the Distribution Generation facility from the Distribution System, in accordance with these Regulations.

11. Voltage and Frequency Range.- A variation of ±5% and ±1% is permissible to the nominal voltage and frequency respectively.

INTERCONNECTION FACILITY FINANCING

12. Responsibility for Costs of Interconnecting a Distributed Generation Facility.— (1) A Distributed Generator shall be responsible for all costs associated with Interconnection Facilities up to the Interconnection Point including metering installation.

(2) The Distributed Generator shall also be responsible for any costs reasonably incurred by Distribution Company in providing, operating, or maintaining Interconnection Facilities and Distribution System improvements required solely for the interconnection of the Distributed Generation Facility with Distribution Company's Distribution System.

(3) In case of non-availability of meter(s) with Distribution Company, the Distributed Generator may procure such meter(s) directly subject to testing by Distribution Company, before installation.

NET ENERGY METERING SERVICES

13. Metering Requirements.— (1) The equipment installed for net metering shall be capable of accurately measuring the flow of electricity in two directions.

Provided that in case two separate meters are installed, the net energy metering calculation shall yield the same result as when a single meter is used.

(2) The Net Energy Metering Facility, shall meet all safety and protection requirements that are necessary to assure safe and reliable operation of the Distributed Generation Facility when connected to the Distribution System of the Distribution Company and that have been approved by the Authority.

14. Billing for Net Metering.— (1) At the end of each Billing Cycle following the date of final interconnection of Distributed Generation Facility to the Distribution System of the Distribution Company, the Distribution Company shall net off the kWh supplied by Distributed Generator against the kWh supplied by it.

Provided that the meter readings shall be carried out preferably through Hand Held Units (HHU) and through automated means as directed by the Authority from time to time.

(2) In case the kWh supplied by Distribution Company exceed the kWh supplied by Distributed Generator, the Distributed Generator shall be billed for the net kWh in accordance with the Applicable Tariff.

(3) In case the kWh supplied by Distributed Generator exceed the kWh supplied by Distribution Company, the net kWh shall be credited against Distributed Generator's next billing cycle for future consumption, or shall be paid by the Distribution Company to the Distributed Generator quarterly.

Provided that where the Distributed Generator is to be paid, the kWh in a month will be charged at the tariff of that respective month.

(4) The tariff payable by the Distribution Company shall only be the off-peak rate of the respective consumer category of the respective month and other rates such as variable charges for peak time, fixed charges, fuel price adjustment, duties/levies will not be payable by Distribution Company.

15. Power of the Authority to give directions, instructions and guidelines.— (1) For carrying out the purposes of these Regulations, the Authority may issue directions, instructions and guidelines to the Distributed Generator and the Distribution Company in the form and manner determined by the Authority, which shall be complied with by the Distributed Generator and the Distribution Company.
(2) The Authority may, on representation made to it or on its own motion modify or cancel any direction, instruction or guidelines issued under sub-rule (1), and in so modifying or cancelling any direction, instruction or guidelines may impose such conditions as it thinks fit.

16. Power to require information.—The Authority may, at any time, by notice in writing require any director, officer and member of the Distribution Company and/or the Distributed Generator, generally or in particular to furnish it within the time specified therein or such further time as the Authority may allow, with any statement or information and without prejudice to the generality of the foregoing power, may call for information, at such intervals as the Authority may deem necessary.

17. Resolution of disputes.— Any dispute or disagreement between Distributed Generator and Distribution Company relating to any matter arising out of, or in connection with, the activities covered under the Regulations shall be submitted for decision to the Authority.

18. Penalty for failure, refusal to comply with, or contravention with any provision of the Regulations.— If any person fails or refuses to comply with, or contravenes any of the provisions of these Regulations or any direction or order passed by the Authority under these Regulations or knowingly or will fully authorizes or permit such failure, refusal or contravention, he shall, be punishable with a fine which may extend to 100 million rupees.

No. NEPRA/NMR-01/

(Syed Safeer Hussain)
Registrar

Annexure

Schedule-I
[See Regulation 2 (1) (b)]

Distributed Generation Interconnection Agreement (1 KW to 1MW) between Distributed Generator and Distribution Company

The Distributed Generation Interconnection Agreement (the "Agreement"), is made and entered in to this _____ (day) of _____ (month), _____ (year) by and between _____ hereinafter called "DISCO" and _____ hereinafter called the "Applicant". Applicant shall be 3 phase 400V or 11kV DISCO's consumer. DISCO and the Applicant are hereinafter collectively referred to as the "Parties" and individually as a "Party".

Recitals

A. DISCO is the owner of the electric distribution system serving _____ [Insert legal description of property or address] "DISCO's Distribution System"

B Applicant desires to install a Distributed Generator (DG) facility or energy storage device using solar or wind energy resources with a capacity greater than 1 KW but no more than 1 MW, including related interconnection equipment (the "DG Facility") and to interconnect the DG Facility to the DISCO's distribution system.

C. DISCO has previously reviewed and approved Applicant's DG Interconnection Applicant Form dated _____, and supporting materials (the Application"). The completed Application is attached as Exhibit 1 and incorporated into this Agreement.

D. Applicant wishes to interconnect the DG Facility to DISCO's distribution system and DISCO is willing to permit such interconnection subject to the terms and conditions set forth: (1) the completed Application approved by DISCO; (2) this Agreement.

E. No agency or partnership is created with the interconnection of the applicants DG Facility.

Agreement

NOW THEREFORE, in consideration of the foregoing Recitals and for good and valuable consideration, the DISCO and Applicant agree as follows:

1. Design Requirement

The DG Facility shall be installed in compliance with NEPRA (Alternative & Renewable Energy) Distributed Generation/ Net Metering Regulations 2015.

2. Applicant's Representations and Warranties

Applicant represents and warrants that:

a. The DG Facility is fully and accurately described in the Application;
b. All information in the Application is true and correct;
c. The DG Facility has been installed to Applicant's satisfaction;
d. Applicant has been given warranty information and an operation manual for the DG Facility;
e. Applicant has been adequately instructed in the operation and maintenance of the DG Facility.

3. Interconnection Disconnect Switch

The DISCO may require that the Applicant furnish and install an interconnection disconnect switch that opens, with a visual break, all underground poles of the interconnection circuit. The interconnection disconnect switch shall be rated for the voltage and fault current requirements of the DG Facility, and shall meet all applicable IEC, IEEE Standards, as well as applicable requirements of the NEPRA Grid Code. The switch enclosure shall be properly grounded. The interconnection disconnect switch shall be accessible at all times, located for ease of access of DISCO personnel, and shall be capable of being locked in the open position. The Applicant shall follow the DISCO's recommended switching, clearance, tagging, and locking procedures.

4. Modifications to the DG Facility

Applicant shall notify DISCO of plans for any material modification to the DG Facility by providing at least forty working days of advance notice. "Material Modification" is defined as any modification that changes the maximum electrical output of the DG Facility or changes the interconnection equipment. The notification shall consist of a completed, revised Application and such supporting materials as may be reasonably requested by DISCO. Applicant agrees not to commence installation of any material modification to the DG Facility until DISCO has approved the revised Application.

5. Insurance, Indemnification, Liability

5.1 Distributed Generator shall obtain and maintain appropriate insurance for third party personal injury and general commercial liability.

5.2 Each party as indeminitor shall defend, hold harmless, and indemnify the other party and the directors, officers, employees, and agents of the other party against and from any and all loss, liability, damage, claim, cost, charge, demand, or expense (including any direct, indirect or consequential loss, liability, damage, claim, cost, charge, demand, or expense, including attorneys' fees) for injury or death to persons, including employees of either party, and damage to property, including property of either party, arising out of or in connection with (a) the engineering, design, construction, maintenance, repair, operation, supervision, inspection, testing, protection or ownership of the indeminitor's facilities, or (b) the making of replacements, additions, betterments to, or reconstruction of the indeminitor's facilities. This indemnity shall apply notwithstanding the active or passive negligence of the indeminitee. However, neither party shall be indemnified hereunder for its loss, liability,

damage, claim, cost, charge, demand, or expense resulting from its sole negligence or willful misconduct.

5.3 The indeminitor shall, on the other party's request, defend any suit asserting a claim covered by this indemnity and shall pay for all costs, including reasonable attorney fees, that may be incurred by the other party in enforcing this indemnity.

5.4 The provisions of this Section shall not be construed to relieve any insurer of its obligations to pay any insurance claims in accordance with the provisions of any valid insurance policy.

5.5 Except as otherwise provided in this section, neither party shall be liable to the other party for consequential or remote damages incurred by that party.

6. DG Facility commissioning Testing

Applicant shall notify DISCO in writing that installation of DG Facility is complete and that the interconnection equipment is available for testing by DISCO at least fifteen working days duly certified by Electrical Inspector before Applicant interconnects the DG Facility with DISCO's Distribution System. DISCO shall thereupon have the right to test the DG Facility. DISCO shall also have the right to witness any testing by Applicant of the DG Facility. Any DISCO testing of the DG Facility shall be completed within ten working days. After the testing which is to the satisfaction of both parties, the DG facility may be interconnected with the distribution system of DISCO to be witnessed by representatives of both parties within thirty eight days.

7. Access to DG Facility

Applicant shall permit DISCO's employees and agents to enter the property on which the DG Facility is located at any reasonable time for the purposes of inspecting and/or testing Applicant's DG Facility to ensure its continued safe and satisfactory operation and the accuracy of DISCO meters. Such inspections shall not relieve Applicant from its obligation to maintain the DG Facility and any related equipment owned by Applicant in safe and satisfactory operating conditions.
DISCO shall have the right to witness any testing by Applicant of the DG Facility.

8. Temporary Disconnection of a DG Facility

DISCO may limit the operation and/or disconnect or require the disconnection of a DG facility from DISCO's Distribution System at any time, with or without notice, in the event of fault. DISCO may also limit the operation and/or disconnect or require the disconnection of DG facility from DISCO's Distribution System upon the provision of 30 days written notice for the conditions to allow for routine maintenance, repairs or modifications to DISCO's Distribution System:

9. Disputes; Right to Appeal to Authority

Nothing in this Agreement prevents Applicant from filing a petition with the Authority to appeal any requirement imposed by DISCO as a condition to interconnection of DG Facility